

DATA ANALYSIS REPORT

Advancing Circular Economy Practices in Agricultural Waste Management Across Selected African Countries

Geographic Scope: Kenya · Nigeria · Cameroon · South Africa · Rwanda

Decision-Support Analysis & Power BI Dashboard

Final Report

Sustainable Development Goals Addressed: SDG 2 · SDG 8 · SDG 12 · SDG 13

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1. Project Overview

This project applies data analytics and visualization techniques to evaluate agricultural waste generation and post-harvest food loss across selected African countries, with the goal of identifying investment-ready circular economy opportunities that support job creation, environmental sustainability, and industrial innovation.

The analysis operationalizes the approved research proposal through a fully developed Power BI decision-support dashboard, transforming raw multi-source datasets into actionable insights for policymakers, investors, and development practitioners.

5 Countries Analyzed	5 Core Data Sources	4 SDGs Aligned	1 Power BI Dashboard
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2. Project Objectives (As Executed)

The completed analysis addressed the following objectives:

- Quantify agricultural waste and post-harvest food loss by country and crop
- Integrate socioeconomic indicators (unemployment, poverty, emissions) with waste data
- Identify regions where circular economy interventions can deliver maximum impact
- Communicate insights through an interactive Power BI dashboard for non-technical users

Status

All objectives outlined in the original proposal were successfully implemented through structured data analysis and visualization.

3. Data Sources and Scope

The analysis combines agricultural, environmental, and socioeconomic datasets sourced from reputable international and national institutions:

Source	Data Provided
FAOSTAT	Agricultural production and food loss data
World Bank	Poverty, population, and development indicators
ILOSTAT	Employment and unemployment statistics
UNEP	Waste and emissions context
National statistical agencies	Country-level agricultural reporting

Geographic Coverage

- Kenya
- Nigeria
- Cameroon
- South Africa
- Rwanda

4. Data Preparation and Processing

4.1 Data Cleaning

The following preprocessing steps were completed before visualization:

- Standardization of measurement units (metric tons, percentages)
- Harmonization of country names and codes across datasets
- Removal of duplicate records
- Handling of missing values through exclusion or aggregation
- Temporal alignment of indicators to comparable reporting years

These steps ensured consistency and analytical validity across all datasets used in Power BI.

5. Analytical Methodology

5.1 Descriptive Analysis

Initial exploratory analysis focused on:

- Total agricultural production volumes
- Food loss and waste quantities by country
- Crop-level contributions to waste generation

This established baseline patterns and identified high-waste-producing regions.

5.2 Socioeconomic Correlation Analysis

Agricultural waste indicators were analyzed alongside:

- Unemployment rates
- Poverty levels
- Greenhouse gas emissions related to food systems

This integration enabled identification of regions where waste mismanagement overlaps with economic vulnerability.

5.3 Comparative Country Analysis

Countries were compared across standardized metrics to reveal:

- Relative waste intensity
- Labor market pressure
- Environmental impact exposure

This comparative approach supports evidence-based prioritization rather than single-country analysis.

6. Power BI Dashboard Implementation

6.1 Dashboard Structure

The Power BI dashboard was designed as a decision-support system, consisting of:

- Country-level overview pages
- Comparative bar and column charts
- KPI cards highlighting key indicators
- Filter controls for country, crop, and indicator selection

All visualizations were built directly from the cleaned datasets provided in the Excel workbook.

6.2 Key Visual Insights

The dashboard enables users to:

- Identify countries with the highest agricultural waste volumes
- Compare unemployment rates against waste availability
- Observe relationships between waste generation and emissions
- Highlight priority regions for circular economy investments

Visual patterns replace complex statistical outputs, making insights accessible to non-technical stakeholders.

7. Results: Final Dashboard KPI Summary

This section presents the final quantitative outputs extracted directly from the completed Power BI dashboard. It reflects the dashboard in its finished state and is intended to complement, not replace, the descriptive and methodological content in Sections 1–6.

7.1 Dashboard Scope (Final Data State)

The final dashboard build expanded coverage beyond the five countries named in the original proposal scope to a total of eight countries — Botswana, Cameroon, Congo (Dem. Rep.), Ghana, Kenya, Nigeria, Rwanda, and South Africa — spanning 22 years of data. The table below summarizes the headline KPI cards from the dashboard’s landing page.

Dashboard KPI (Final State)	Value
Countries Analyzed (Final)	8
Years of Data Covered	22
Total Population Represented	9bn (cumulative, country-year records)
Total Agricultural Production (index)	15.64K
General Average % Food Loss	3.43%
General Average % Rate of Unemployment	9.61%
Average Gas Emissions	162.64K

7.2 Country-Level Findings

- **Gas emissions are heavily concentrated in two countries.** Congo (Dem. Rep.) records the highest average gas emissions ($\approx 0.59\text{M}$), followed by Nigeria ($\approx 0.39\text{M}$) — together far exceeding all other countries in the dataset, where emissions cluster below 0.1M .
- **Food loss shares are highest in Ghana.** Ghana shows the highest average % food loss ($\approx 5\%$), followed by Kenya, Rwanda, Cameroon, and South Africa ($\approx 4\text{--}4.5\%$); Botswana records the lowest share ($\approx 3.2\%$).
- **Production volume does not explain the gap.** Average agricultural production is comparably high across most of the eight countries (roughly 75–98 on the dashboard’s production index), while average % food loss varies more widely across the same set — confirming that post-harvest handling and infrastructure, not production volume, drive the disparity.

7.3 Supply Chain and Commodity-Level Findings

- **Post-harvest is the costliest supply chain stage.** The post-harvest stage accounts for the largest share of average % food loss ($\approx 26\%$), more than $1.5\times$ the next-highest stage, retail ($\approx 15\%$), with wholesale and whole-supply-chain stages following.
- **Storage drives the largest cumulative losses by activity.** Storage-related activities show the highest sum of % food loss ($\approx 3.8\text{K}$), ahead of drying/harvesting ($\approx 2.5\text{K}$) and transportation ($\approx 1.9\text{K}$), pointing to storage infrastructure as the leading intervention point.
- **Losses are concentrated in a small set of perishable commodities.** Papayas record the highest average % food loss ($\approx 48\%$), followed by watermelons, strawberries, cattle, and other tropical fruits — all well above the grain and legume commodities at the bottom of the ranking.

7.4 Socioeconomic and Environmental Correlation

- **Unemployment has trended upward while food loss has fluctuated.** Average % rate of unemployment rose from roughly 8–9% in the early 2000s to above 12% in the most recent year, while average % food loss fluctuated within a narrower 2–5% band over the same period, with the two series converging after 2015.
- **High-loss, high-emission countries cluster geographically.** The regional view shows that countries combining high average % food loss with high gas emissions and elevated unemployment sit predominantly in Central and West Africa, reinforcing the dashboard’s built-in insight that these regions represent the strongest opportunities for investment in storage, logistics, food processing, and recycling industries.

8. Key Findings

8.1 Waste Concentration Is Uneven

Agricultural waste generation varies significantly between countries, even where production volumes are similar. This indicates inefficiencies in post-harvest handling, storage, and processing rather than production alone.

8.2 Strong Alignment Between Waste Availability and Job Creation Potential

Countries with high agricultural waste volumes also show elevated unemployment and poverty indicators, signaling strong potential for labor-intensive circular economy interventions.

8.3 Environmental Impact Can Be Mitigated Through Waste Conversion

Food system-related emissions are closely linked to unmanaged waste. Diverting organic waste into controlled conversion pathways (biogas, feed, compost) presents clear climate mitigation opportunities.

9. Policy and Investment Implications

The completed analysis supports the following recommendations:

- Prioritize waste-to-value investments in high-waste, high-unemployment regions
- Use the Power BI dashboard as an ongoing monitoring tool for policy planning
- Support pilot projects that combine waste management with green job creation
- Expand the analytical framework to additional African countries

10. Alignment With Sustainable Development Goals

SDG	Focus Area
SDG 8	Decent Work and Economic Growth
SDG 2	Zero Hunger
SDG 12	Responsible Consumption and Production

SDG 13	Climate Action
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11. Project Deliverables (Completed)

- Cleaned and harmonized datasets
- Fully functional Power BI dashboard
- Data-driven country comparisons
- Actionable insights for policymakers and investors
- Reproducible analytical framework

12. Conclusion

This project demonstrates how data analytics and visualization can translate complex agricultural and socioeconomic datasets into practical decision-support tools. By integrating waste, employment, and environmental indicators into a single analytical framework, the study provides a scalable model for advancing circular economy strategies across Africa.

The Power BI dashboard serves as the final analytical artifact, enabling evidence-based policy formulation and investment prioritization.

13. Limitations and Future Work

- Data availability varied across countries and years
- Sub-national (regional) data was limited in some cases
- Future work could integrate real-time data and predictive modeling